

Experiment Title

Multiplication of Two Numbers Using 8086 Assembly Language

Objective

The aim of this experiment is to learn how multiplication is performed in the Intel 8086 microprocessor using assembly language instructions. This lab focuses on understanding register usage and the MUL instruction.

Required Tools

- Intel 8086 Processor
- Assembly Language
- Emulator (EMU8086 or equivalent)
- PC with Windows Operating System

Theory

The Intel 8086 supports unsigned multiplication through the MUL instruction. When an 8-bit value is multiplied, the processor assumes one operand is stored in the **AL** register, and the result is automatically placed in the **AX** register. Proper understanding of register roles is essential for correct execution.

Procedure

1. Load the data segment into the DS register.
2. Store the first number in the AL register.
3. Store the second number in another general-purpose register.
4. Execute the multiplication instruction.
5. Observe the result in the AX register.

Program Code

```
.model small
.stack 100h
.data
    num1 db 5
    num2 db 3
    msg db "Result = $"
.code
main proc
    mov ax, @data
    mov ds, ax
```

; multiply

mov al, num1

mov bl, num2

mul bl ; AX = result

mov bx, ax ; save result safely

; print message

mov dx, offset msg

mov ah, 09h

int 21h

; convert number in BX to decimal

mov ax, bx

mov cx, 0

mov bx, 10

convert:

xor dx, dx

div bx

push dx

inc cx

test ax, ax

jnz convert

print:

pop dx

add dl, 30h

mov ah, 02h

```
int 21h
```

```
loop print
```

```
mov ah, 4ch
```

```
int 21h
```

```
main endp
```

```
end main
```

Output



The product of the two numbers is successfully generated and stored in the AX register.

Conclusion

This experiment demonstrates how multiplication works at the assembly language level in the 8086 microprocessor. It provides a clear understanding of register dependency and arithmetic instruction execution.